

TURAIF, Saudi Arabia

WMO: 403560

Lat: 31.693N

Lon: 38.731E

Elev: 854

StdP: 91.47

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.8	-12.2	1.5	11.7	-10.8	1.7	12.5	12.1	9.1	10.8	8.8	3.1	100	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.2	40.9	18.8	39.2	18.3	38.0	17.7	20.1	35.1	19.4	34.5	18.8	33.9	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.7	23.0	15.0	11.8	22.6	13.9	11.0	22.5	61.2	34.6	58.8	34.6	56.7	33.2	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.4	8.3	DB	-4.0	42.7	1.7	1.0	-5.2	43.4	-6.2	44.0	-7.2	44.6	-8.4	45.3	(4)
(5)			WB	-5.3	21.2	1.8	1.2	-6.6	22.0	-7.7	22.7	-8.7	23.3	-10.0	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	7.7	9.8	13.8	18.6	23.9	27.5	29.6	29.9	26.9	21.9	14.1	9.3	(6)	
	DBStd	8.52	2.87	3.31	3.83	3.97	3.40	2.50	2.41	2.47	2.63	3.26	3.56	3.04	(7)	
	HDD10.0	183	81	40	7	1	0	0	0	0	0	0	7	48	(8)	
	HDD18.3	1186	328	241	149	45	3	0	0	0	0	9	132	279	(9)	
	CDD10.0	3639	11	33	126	258	429	524	608	617	507	368	130	27	(10)	
	CDD18.3	1599	0	1	9	53	174	274	350	358	257	118	6	0	(11)	
	CDH23.3	22286	0	9	144	663	2211	3895	5279	5346	3363	1293	80	2	(12)	
	CDH26.7	12882	0	1	39	228	1081	2304	3359	3421	1908	529	11	0	(13)	
(14) Wind	WSAvg	4.3	4.1	4.4	4.7	4.7	4.7	4.5	4.5	4.2	4.0	3.9	3.8	3.7	(14)	
(15) Precipitation	PrecAvg	53	5	12	4	3	3	0	0	0	1	7	6	12	(15)	
	PrecMax	138	29	39	35	19	28	2	1	0	9	95	35	37	(16)	
	PrecMin	23	0	1	0	0	0	0	0	0	1	0	0	1	(17)	
	PrecStd	27	8	11	9	5	6	1	0	0	2	18	9	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	24.9	30.9	33.8	37.2	40.8	42.6	42.2	40.1	35.1	28.1	22.8	(19)	
		MCWB	10.3	12.2	13.8	14.9	17.1	17.9	19.3	19.3	18.6	17.0	13.8	11.9	(20)	
	2%	DB	17.9	21.2	27.1	31.2	35.8	38.8	40.9	41.0	38.0	33.1	25.8	19.9	(21)	
		MCWB	9.4	10.6	12.6	14.1	16.2	17.6	18.8	18.9	17.8	15.9	13.3	10.5	(22)	
	5%	DB	15.8	19.0	24.1	29.2	33.9	37.0	39.1	39.2	36.2	31.8	23.2	17.2	(23)	
		MCWB	8.7	9.5	11.5	13.4	15.6	16.7	18.3	18.4	17.2	15.7	12.6	9.8	(24)	
	10%	DB	13.8	16.8	22.0	27.2	32.1	35.8	37.8	38.0	34.8	29.9	21.2	15.2	(25)	
		MCWB	8.0	8.6	10.8	12.8	14.8	16.3	17.6	17.9	16.9	15.5	12.0	9.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	13.7	15.6	16.9	18.3	19.6	20.7	21.2	20.1	18.9	16.3	13.8	(27)	
		MCDB	17.2	21.3	24.6	27.3	33.1	35.0	36.4	35.4	34.0	30.7	22.4	18.6	(28)	
	2%	WB	10.2	11.6	13.6	15.5	17.3	18.6	19.7	20.3	19.1	17.6	15.1	12.1	(29)	
		MCDB	15.2	18.5	23.4	26.3	31.9	34.4	35.5	35.4	33.4	28.6	21.4	17.1	(30)	
	5%	WB	9.2	10.4	12.4	14.5	16.5	17.7	19.1	19.6	18.4	16.8	14.0	10.8	(31)	
		MCDB	14.3	16.9	22.6	26.5	30.6	33.5	35.1	34.6	32.6	27.8	20.2	15.7	(32)	
	10%	WB	8.4	9.3	11.3	13.5	15.7	17.2	18.6	18.9	17.7	16.1	13.1	9.7	(33)	
		MCDB	13.1	15.4	20.7	25.4	29.6	32.8	34.2	33.9	31.6	26.6	19.1	14.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.7	12.9	13.5	14.1	15.1	15.2	15.2	14.8	13.8	12.3	11.2	(35)	
		MCDBR	14.0	15.3	15.9	15.5	15.5	16.3	16.1	16.0	15.7	15.4	14.4	13.7	(36)	
	5% WB	MCWBR	7.4	7.4	7.4	6.5	5.9	5.7	5.2	5.2	5.8	6.3	6.9	7.5	(37)	
		MCWBR	11.9	12.9	14.4	13.9	14.1	15.2	15.3	15.4	14.7	13.4	11.7	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.329	0.357	0.406	0.475	0.479	0.407	0.402	0.409	0.396	0.426	0.367	0.337	(40)	
		taud	2.269	2.174	2.015	1.841	1.857	2.076	2.116	2.097	2.115	2.015	2.158	2.245	(41)	
	Ebn at Noon		884	891	870	822	821	878	881	871	868	802	827	850	(42)	
		Edh at Noon	109	132	167	208	206	165	158	159	149	154	121	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.50	4.39	5.69	6.69	7.55	8.36	8.04	7.37	6.38	4.93	3.80	3.24	(44)	
	RadStd		0.16	0.28	0.21	0.21	0.33	0.13	0.19	0.16	0.21	0.22	0.24	0.13	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	+1.04	+1.28	+1.58	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.36	N/A	N/A	+0.59	+1.01	+1.06	N/A	N/A	+125

Nomenclature: See separate page